ORDINARY DIFFERENTIAL EQUATIONS • UNIT 4.01

C Nonhomogeneous Equations, Undetermined Coefficients & Resonance

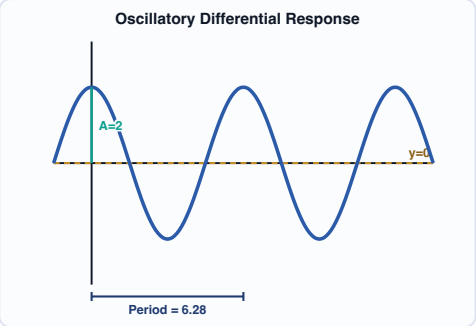
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Nonhomogeneous Equations, Undetermined Coefficients & Resonance Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 39

B 24

C 42

D 53

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

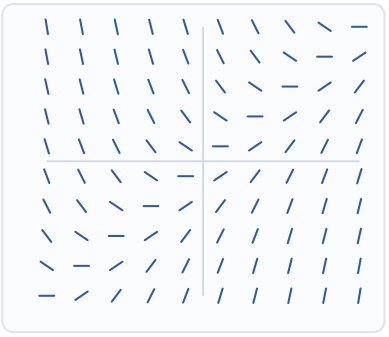
B $e^2 - 1$

C $3 - 2e^{-2}$

D $3 - 2e^{-1}$

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D 1.220

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

A 5

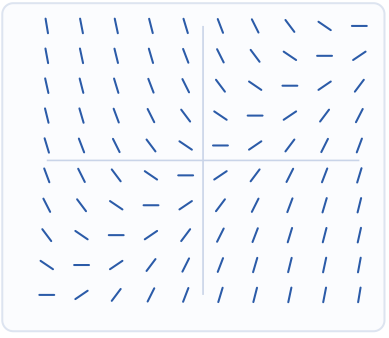
B -5

C 6

D -6

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



A 7

B 6

C 8

D 5

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

A $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

B $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

C $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

D $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 44

B 43

C 24

D 40

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

A 170

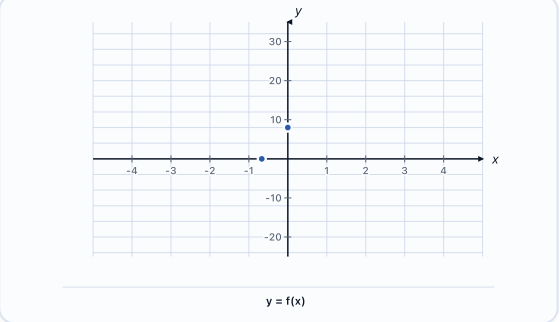
B 151

C 173

D 187

9 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1 e^{2x} + c_2 e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.



A $A = 3$

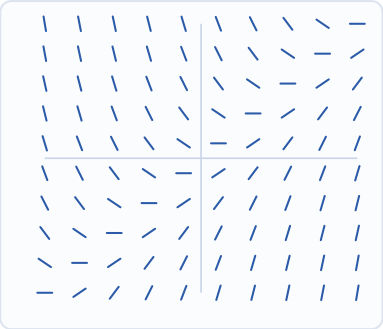
B $A = 2$

C $A = 6$

D $A = 12$

10 MCQ

The first-order differential equation $\frac{dF}{dw} = 16w - 3F$ has its slope field plotted below. Determine the slope $\frac{dF}{dw}$ at the point $D(18, 8)$.



A 280

B 267

C 256

D 264

11 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

A 2

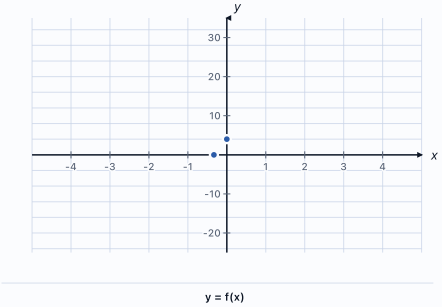
B $2x$

C $2\sin(2x)$

D $2\cos(2x)$

12 MCQ

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.



A 1

B 6

C 12

D 2

13 MCQ

Consider the second-order linear nonhomogeneous differential equation: $y''(x) - 3y'(x) + 2y(x) = 4$. Find the particular solution, $y_p(x)$, using the method of Undetermined Coefficients.

A $y_p(x) = 2$

B $y_p(x) = 2x$

C $y_p(x) = 4$

D $y_p(x) = 4x$

14 MCQ

The first-order differential equation $\frac{df}{dx} = 19x - 4f$ has its slope field plotted below. Find the slope $\frac{df}{dx}$ at the point $A(5, 5)$.

A 78

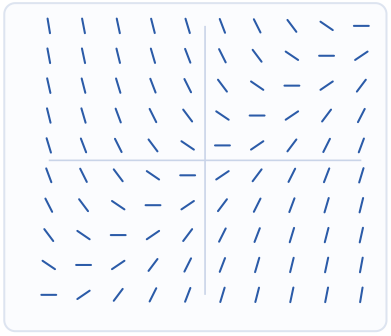
B 75

C 94

D 70

15 MCQ

Solve $y'' + y = 3\cos 2x$, $y(0) = 2$, $y'(0) = 0$. Find $y(\pi)$.



A -4

B 2

C 4

D -2

16 MCQ

Solve $y'' - 4y = 8x$, $y(0) = 0$, $y'(0) = 2$. Find $y(1)$.

A $e^2 - e^{-2} - 2$

B $2e^2 - 2$

C $e^2 + e^{-2} - 2$

D $e^2 - e^{-2} + 2$

17 MCQ

Solve $y'' - y' - 2y = 3e^{2x}$, $y(0) = 1$, $y'(0) = 6$. Find $y(1)$.

A $3e^2 + e^{-1}$

B $3e^2 - e^{-1}$

C $e^2 - e^{-1}$

D $3e^2 - e$

18 MCQ

Solve $y'' + 9y = 2\sin 3t$, $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/3)$.

A $-\pi/9$

B $\pi/9$

C $\pi/3$

D $2\pi/9$

19 MCQ

Find a particular solution of $y'' - 4y' + 3y = 6x + 4$.

A $2x - 4$

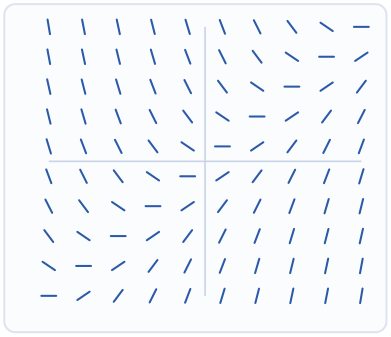
B $2x + 2$

C $2x + 4$

D $4x + 4$

20 MCQ

Solve $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 0$, $y'(0) = 0$. Find $y(2)$.



A $2e^4$

B e^{-4}

C $4e^{-4}$

D $2e^{-4}$

21 MCQ

An undamped oscillator satisfies $y'' + 4y = 3\cos 2t$ (forcing at exactly the natural frequency), $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/4)$.

A $-3\pi/16$

B $3\pi/8$

C $3\pi/16$

D $\pi/4$

22 MCQ

Find a particular solution of $y'' - y' - 6y = e^{2x} + 18$.

A $\frac{1}{4}e^{2x} - 3$

B $-\frac{1}{4}xe^{2x} - 3$

C $-\frac{1}{4}e^{2x} + 3$

D $-\frac{1}{4}e^{2x} - 3$

23 MCQ

Solve $y'' - 2y' - 3y = -3$, $y(0) = 1$, $y'(0) = -8$. Find $y(1)$.

A $-1 - \frac{2}{e} + 2e^3$

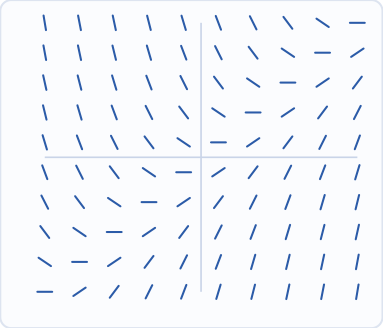
B $-2e^3 + \frac{2}{e} + 1$

C $-4e^3 + \frac{4}{e} + 2$

D $-e^3 + e^{-1} + \frac{1}{2}$

24 MCQ

Calculate the solution set for $y'' + 3y' + 2y = 2$, $y(0) = 0$, $y'(0) = 3$. Find $y(1)$.



A $-\frac{2}{e^2} + e^{-1} + 1$

B $-\frac{4}{e^2} + \frac{2}{e} + 2$

C $-\frac{1}{e^2} + \frac{1}{2e} + \frac{1}{2}$

D $-1 - e^{-1} + \frac{2}{e^2}$

25 MCQ

Solve $y'' - 4y = 4$, $y(0) = 2$, $y'(0) = 2$. Find $y(1)$.

A $-1 + e^{-2} + 2e^2$

B $-2e^2 - e^{-2} + 1$

C $-2 + \frac{2}{e^2} + 4e^2$

D $-\frac{1}{2} + \frac{1}{2e^2} + e^2$